

Two-Step Equations: Scaffolded Practice (Problems 2-4)

Name: _____ Date: _____ Period: _____

Follow each step and fill in the blanks. Each step builds on the one before it.

Problem 2: Use the bar diagram

Preston uses a bar diagram to represent $4x - 3 = 13$. Use it to solve for x .

Total			
13			3
x	x	x	x

Step 1 — Read the diagram. The bottom bar has _____ equal parts, and each part is labeled _____. Together they show the expression _____.

Step 2 — Find the total. The top bar shows 13 plus _____. So the whole bar equals $13 + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$.

Step 3 — Connect to the equation. Adding 3 to the 13 is the same as adding _____ to **both sides** of $4x - 3 = 13$.

$$4x - 3 + \underline{\hspace{2cm}} = 13 + \underline{\hspace{2cm}}$$

$$4x = \underline{\hspace{2cm}}$$

Step 4 — Split the total into equal parts. The total is shared equally among _____ parts, so _____ both sides by _____.

$$x = \underline{\hspace{2cm}} \div \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

Step 5 — Check. $4(\underline{\hspace{2cm}}) - 3 = \underline{\hspace{2cm}} - 3 = \underline{\hspace{2cm}}$ ✓

Explain: How does the bar diagram help you solve for x ?

Problem 3: Check Clara's answer

Clara solved $6p - 12 = 72$ and says $p = 14$. Is she correct?

Step 1 — Plan. To check a solution, _____ (substitute / subtract) her answer for the variable in the **original** equation.

Step 2 — Substitute. Replace p with _____.

$$6(\text{_____}) - 12 = 72$$

Step 3 — Multiply first. $6 \times \text{_____} = \text{_____}$

$$\text{_____} - 12 = 72$$

Step 4 — Subtract. _____ $- 12 = \text{_____}$

Step 5 — Compare. Is the left side equal to the right side? _____ (yes / no)

_____ $= 72$ is a _____ (true / false) statement.

Step 6 — Conclude. Clara _____ (is / is not) correct, because

Try it another way: Solve $6p - 12 = 72$ yourself.

$$6p - 12 + \text{_____} = 72 + \text{_____} \rightarrow 6p = \text{_____} \rightarrow p =$$

$$\text{_____} \div 6 = \text{_____}$$

Problem 4: Clyde's flour

The recipe needs **$1\frac{1}{3}$ cups** of flour. Clyde has **2 cups**, but he is **doubling** the recipe. How much more flour does he need? Let **c** = the amount of flour Clyde still needs.

Step 1 — Pick out the facts.

- Flour in one recipe: _____ cups
- Number of recipes Clyde is making: _____
- Flour Clyde already has: _____ cups

Step 2 — Find the total flour needed. Doubling means multiply by _____.

Total needed = $2 \times 1\frac{1}{3} = 2 \times$ _____ (improper fraction) = _____ =
_____ cups (mixed number)

Part a — Write the equation. What Clyde has + what he still needs = total needed.

$$\text{_____} + c = \text{_____} \times \text{_____}$$

Part b — Solve the equation.

Step 3 — Simplify the right side. $2 + c =$ _____

Step 4 — Undo the addition. _____ (add / subtract) 2 on both sides.

$$2 + c - \text{_____} = \text{_____} - \text{_____}$$

$$c = \text{_____} \text{ cup}$$

Step 5 — Check. $2 +$ _____ = _____, which matches the total needed ✓

Answer in a sentence: Clyde needs _____ cup more flour.